ASSIGNMENT 1 — Q6

Progressive Discount System

JavaScript solution

```
// Q6: Progressive Discount System
"use strict";

const purchaseTotal = 7500; // Replace with the purchase amount.
let discountRate;

if (purchaseTotal >= 10000) {
    discountRate = 0.25;
} else if (purchaseTotal >= 5000) {
    discountRate = 0.15;
} else if (purchaseTotal >= 2000) {
    discountRate = 0.05;
} else {
    discountRate = 0;
}

const discountAmount = Math.round(purchaseTotal * discountRate);
const finalPrice = Math.round(purchaseTotal - discountAmount);

console.log(`Original total: ■${Math.round(purchaseTotal)}`);
console.log(`Discount: ${discountRate * 100}%`);
console.log(`Final price: ■${finalPrice}`);
```

Logic: Selects the highest applicable discount tier and uses Math.round() for the discount and final amount.

How to test: Save this code as a .js file and run it with Node.js using `node filename.js`, or paste it into the browser console.